

A → B → C

ACTIVITY-4

$$h(x) = x \bmod 10$$

a) $4371 \div 10 \rightarrow \text{index } 1$

$1323 \div 10 \rightarrow \text{index } 3$

$6173 \div 10 \rightarrow \text{index } 3 \rightarrow \text{collision.}$

$4199 \div 10 \rightarrow \text{index } 9$

$4344 \div 10 \rightarrow \text{index } 4$

$9679 \div 10 \rightarrow \text{index } 9 \rightarrow \text{collision.}$

$1989 \div 10 \rightarrow \text{index } 9 \rightarrow \text{collision.}$

Result:

Index Chain

0

1

2

3

4

5

6

7

8

9

4371

1323 → 6173

4344

4199 → 9679 → 1989

b) Formula: $h_i(x) = (h(x) + i) \bmod 10$

$i = 0, 1, 2, \dots$

Insertion:

4371: 1 → index 1

1323: 3 → index 3.

6173: 3 → occupied → 4 → index 4.

mixtures
wble

4199: 9 → index 9

1344: 4 → occupied → 0 → index 0

1989: 9 → occupied → 0 occupied → 1 occupied → 2 → index 2

c) $h_1(x) = (h(x) + i^2) \bmod 10$

1371: 1 → index 1

1323: 3 → index 3

6173: 3 → occupied

$i = 1: 3 + 1^2 = 4 \rightarrow \text{index } 4$

4199: 9 → index 9

4344: 4 occupied

$i = 1: 4 + 1 = 5 \rightarrow \text{index } 5$

9679: 9 occupied

$i = 1: 9 + 1 = 0 \rightarrow \text{index } 0$

1989: 9 occupied

$i = 1: 9 + 1 = 0 \rightarrow \text{occupied}$

$i = 2: 9 + 4 = 3$

$i = 3: 9 + 9 = 8 \rightarrow \text{index } 8 \rightarrow \text{free}$

d) $h_2(x) = 7 - (x \bmod 7)$

$h_i(x) = (h_1(x) + i \cdot h_2(x)) \bmod 10$

h_2 : key	$h_1 = x \bmod 10$	$x \bmod 7$
1371	1	1
1323	3	0
6173	3	4
4199	9	4
4344	4	4
9679	9	0
1989	9	6

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